

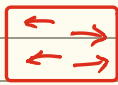
CCNA
Journey

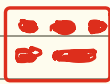
WEEK 1

Day 1: Network Devices

A Computer Network is a digital telecommunications network which allows nodes to share resources.

Router 

Switch 

Firewall 

Server  is a device that provides functions or services for clients.

Client  is a device that accesses a service made available by a server.

The Internet 

Switches are used to forward traffic within a LAN (Local Area Network)

⊗ Switches :

- have many network interfaces / ports for end hosts to connect to (usually 24+)
- Provide connectivity to hosts within the same LAN
- do not provide connectivity between LANs / over the ethernet.

* Routers: have fewer network interfaces than switches.
Are used to provide connectivity between LANs
are used to send data over the internet.

* Firewalls:

- Monitor and control network traffic based on configured rules
- can be placed "inside the network," or "outside the network"
- are known as "Next Generation firewalls" when they include more modern and advanced filtering capabilities.

Network firewall: are hardware that filter traffic between networks.

Host-Based firewalls: are software applications that filter traffic entering and exiting a host machine, like a PC

A client: a device that accesses a service made available by a server

A server: a device that provides functions or services for clients

The same device can be a client in some situations, and a server in other situations.

BGP (Border Gateway Protocol)

DNS. (Domain Name System)

Day 2 : Interfaces and Cables

Rj 45 connector : Registered Jack

Ethernet : collection of network protocols (standards.)

Why do we need network protocols / standards?

Network protocols : industry standards that all vendors follow, both in terms of physical standards like connectors and cables, as well like logical standards like IP, the internet protocol.

Bits and Bytes :

Bits : 0 and 1 (Binary System)

Bytes : 8bits

Speed is measure in bit / second (Kbps, Mbps, Gbps) and not bytes per seconds.

1 Kilobit (Kb) = 1,000 bits

1 megabit (Mb) = 1,000,000 bits

1 gigabit (Gb) = 1,000,000,000 bits

1 Terabit (Tb) = 1,000,000,000,000 bits

petabit, exabit, zettabit, yottabit

Ethernet Standard (Copper)

Defined in the IEEE 802.3 standard in 1983

Speed	Common Name	IEEE Standard	Informal Name	Max length
10mps	Ethernet	802.3i	10BASE-T	100m
100mps	Fast Ethernet	802.3u	100BASE-T	100m
1Gps	Gigabit Ethernet	802.3ab	1000BASE-T	100m
10Gps	10 Gig Ethernet	802.3an	10GBASE-T	100m

BASE = Refers to baseband signaling

T = twisted pair

UTP cables : Unshielded Twisted Pair

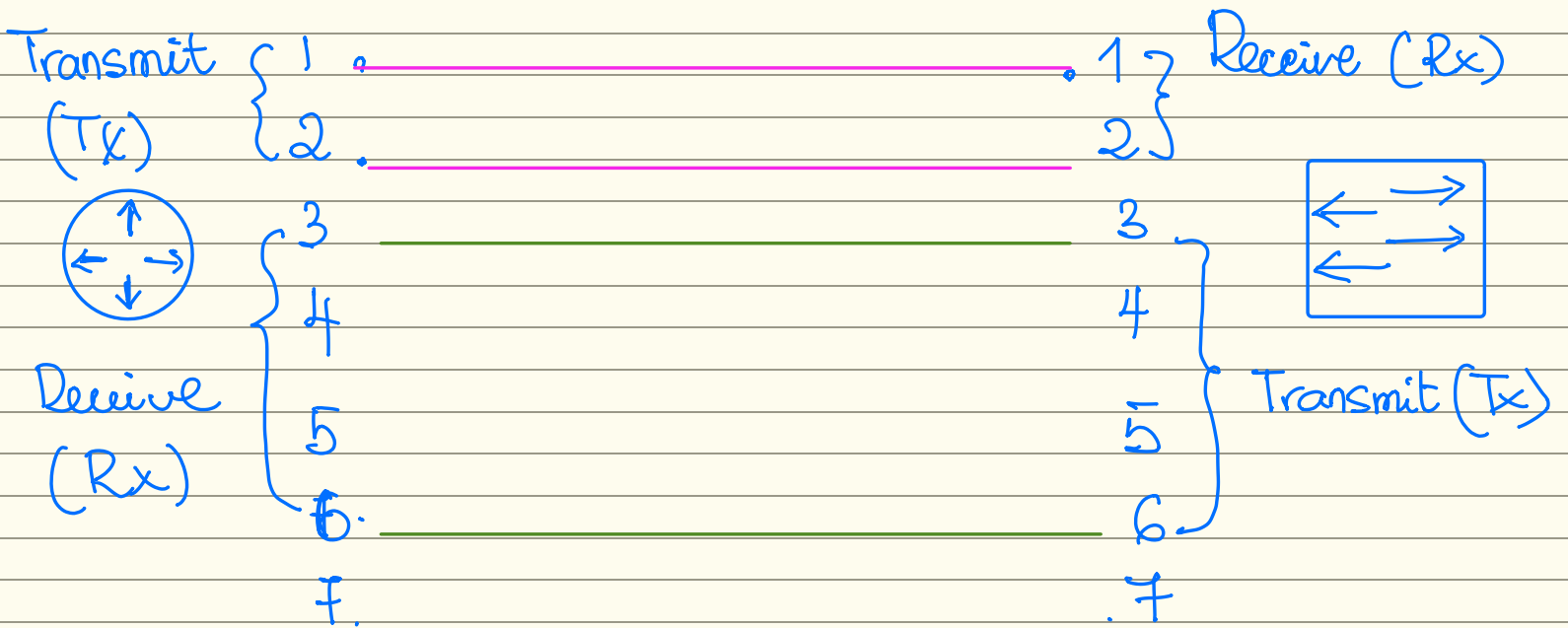
↑
No metallic shield

↑
Protects against EMI
(Electromagnetic Interference)

10 BASE-T
100 BASE-T } 2 pairs (4 wires)

1000 BASE-T
10 G BASE-T } 4 pairs (8 wires)

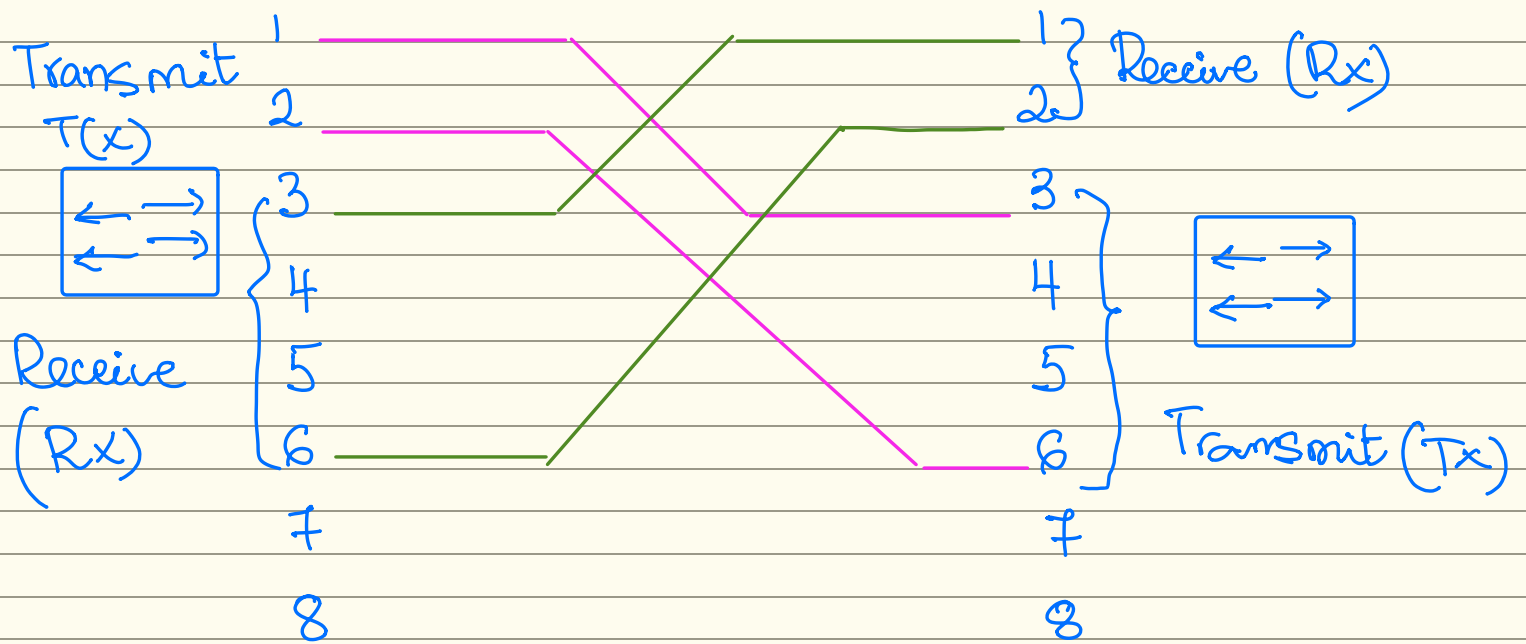
Full - Duplex Trans.



📌 PC transmit data on pins 1 & 2 and Receive data on pins 3 & 6. (Same as Routers)

Switch Receive data on pins 1 & 2 and transmit data on pins 3 & 6.

Router to Router : Crossover Cable .



Router to Router , Switch to Switch , PC to PC or Router to PC .

Device Type	Transmit Pins (Tx)	Receive Pins (Rx)
Router	1 and 2	3 and 6
Firewall	1 and 2	3 and 6
PC	1 and 2	3 and 6
Switch	3 and 6	1 and 2

New networking devices include Auto MDI-X feature. 💡

Crossover Cables : R2R, SRS, P2P, R2P
 Straight Through Cables : P2S, R2S

1000 BASE-T and 10GBASE-T uses all 8 pins, grouped into 4 pairs. (Each pair is bidirectional)

SFP Transceiver (Small Form-Factor Pluggable)

Fiber Optic Cable.

1. Fiberglass core itself
2. cladding that reflects light
3. a protective buffer
4. The outer jacket of the cable

Single Mode fiber and Multimode fiber.

Informal Name	IEEE Standard	Speed	Cable Type	Maximum Length
1000Base-Lx	802.3 z	1Gbps	Multimode or Single-Mode	550 m (MM) 5 km
100GBase-SR	802.3 ae	10Gbps	Multimode	400 m
10GBase-LR	802.3 ae	10Gbps	Single-Mode	10Km
10GBase-LR	802.3 ae	10Gbps	Single-Mode	30Km

Single Mode is more expensive than Multimode fiber
Multimode is often used for shorter distances

UTP cables

- Lower cost than fiber-optic
- Shorter maximum distance than fiber-optic ($\sim 100\text{m}$)
- Can be vulnerable to EMI
- RJ45 ports used with UTP are cheaper than SFP ports
- Emit (leak) a faint signal outside of the cable, which can be copied (= security Risk)

Fiber cables

- Higher cost than UTP
- Longer maximum distance than UTP
- No vulnerability to EMI
- SFP ports are more expensive than RJ45 ports
(Single Mode is more expensive than multimode)
- Does not emit any signal outside of the cable (= No security Risk)

Day 3: How TCP/IP Actually works.

TCP/IP model: a way of grouping protocols into layers based on the job they do.

Application layer	Telnet FTP TFTP HTTP
Transport Layer	TCP UDP
Internet Layer	Internet Protocol & ICMP (IPv4/IPv6)
Link Layer	Local Network Protocol (Ethernet, Wifi)
Physical layer	

Protocol: is a set of rules defining how data should be communicated between devices over a network.

Languages that computers use to communicate.

A standard: is an agreed-upon specification that describes how a protocol or technology should work.

Vin Cerf & Bob Kahn (DARPA) → TCP in 1974

TCP: Transmission Control Program.
Protocol

IP : Internet Protocol

IEEE : Develops many of the technologies used on LAN
Ex. Ethernet, Wifi

IETF (Internet Engineering Task Force)

- Open community that defines protocols used on the Internet (TCP, IP, UDP, HTTP, DNS, etc)

Protocols live (mostly) at one layer

L5/L7

Application Layer

• Protocols for communication between application processes, create & interpret data

L4

Transport Layer

• Provides end-to-end communication between application processes using port numbers

L3

Internet Layer

• Provides end-to-end communication between hosts across networks using IP addresses & Routers

L2

Local Network Layer

• Provides hop-to-hop delivery within a local Network using MAC Address & Switches

L1

Physical Layer

• Sends bits as electrical, optical, or Radio signals over the physical medium.

MAC (Media Access Control) addresses : identify interfaces.

At each step in the encapsulation / decapsulation process, there is a name given to the message:

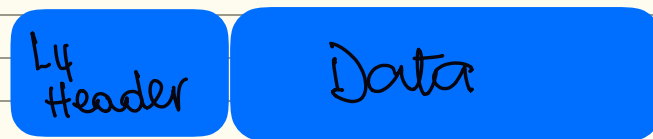
- The combination of data & L4 header is called a segment (TCP) or datagram (UDP)

TCP creates segments, UDP creates datagrams.

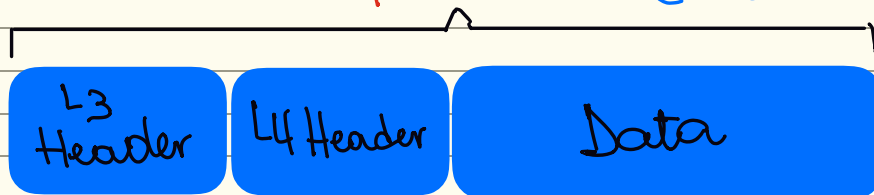
- The combination of a Segment / datagram and a L3 header is called a packet.

- The combination of a packet and a L2 header/trailer is called a frame.

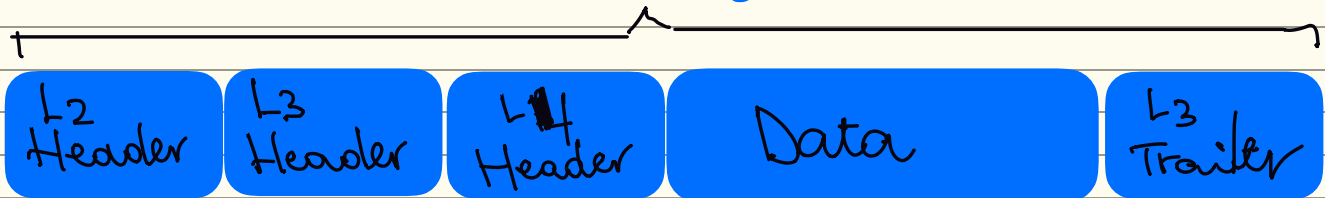
Segment or datagram (L4 PDU)



packet (L3 PDU)



frame (L2 PDU)



PDU: Protocol Data Unit

The contents of each PDU (everything encapsulated by that layer's header / trailer) are called **Payload**.

Payload: What's inside the PDU not including that layer header & trailer.

Same Layer Interaction: Each layer communicates with the same layer on other devices

As long as each layer keeps its contract with the other layers, we can improve or replace protocols at different layers without redesigning everything.

The OSI Model:

International Organization for Standardization (ISO) designed a 7 layers Open System Interconnection (**OSI**) model and a matching protocol suite.

7-layer model

7	Application
4	Transport
3	Network
2	Data Link
1	Physical

OSI model

7	Application
6	Presentation
5	Session
4	Transport
3	Network
2	Data Link
1	Physical

* **STP**: Spanning Tree Protocol (Layer 2)

* **OSPF**: Open Shortest Path First (Layer 3) $\Rightarrow$ used to discover the best paths to different networks.

Day 4 : Intro to CLI

CLI: Command Line Interface

The interface you use to configure Cisco devices

GUI: Graphical User Interface

How to connect to a Cisco device? (console port)

Connecting via Console port.

Rollover Cable: One end → RJ-45; other end → DB9 connector.

Access CLI: Need Terminal Emulator → PuTTY

* Cisco Default:

- Speed of 9600 bits/s; - 8 data bits; 1 Stop bit; no parity, and no flow control.

* **User EXEC Mode** when first enter the CLI (default)

- User EXEC Mode very limited

- User can look at some things, but can't make any changes to the configuration.

* **privileged EXEC Mode**:

Router > enable

Router #

- Provides complete access to view the device, etc.

- Cannot change the configuration, but can change the time on the device, save the configuration.

* **Global Configuration Mode**

Router # configure terminal
Enable password

running - config / Startup - config.

There are two separate configuration files kept on the device at once.

Running - config = the current, active configuration file on the device.

As you enter commands in the CLI, you edit the active configuration.

Startup - config = the configuration file that will be loaded upon restart of the device.

If you enable **service password - encryption**...

- current password will be encrypted
- future passwords will be encrypted
- the **enable secret** will not be effected

If you disable **service password - encryption**...

- current passwords will not be decrypted
- future passwords will not be encrypted
- the **enable secret** will not be effected

Summary:

Router > = user EXEC mode

Router # = privileged EXEC mode

Router (config #) = global config mode

Router > enable

Router # configure terminal (conf t)

Router (config) # enable password password.

Service password - encryption doesn't change the password itself, only how it is displayed in the configuration

The **enable password** command configures a plain-text password, which is not secure as it can be easily read.

Secure password - encryption encrypts the enable password to make it more secure, but it is a weak form of encryption

The **enable secret** command configures a password that is automatically encrypted. It uses MD5, a more secure form of encryption than the **Service password - encryption** command.

Day 4 Lab : Basic Device Security

1. Change the hostnames of the router and switch to the appropriate names (R1, SW1)
##Use the 'hostname' command in global configuration mode##
2. Configure an unencrypted enable password of 'CCNA' on both devices
3. Exit back to user EXEC mode and test the password
4. View the password in the running configuration
5. Ensure that the current password, and all future passwords, are encrypted
6. View the password in the running configuration
7. Configure a more secure, encrypted enable password of 'Cisco' on both devices
8. Exit back to user EXEC mode and then return to privileged EXEC mode.
Which password do you have to use?
9. View the passwords in the running configuration.
What encryption type number is used for the encrypted 'enable password'?
What encryption type number is used for the encrypted 'enable secret'?
10. Save the running configuration to the startup configuration



Day 5: Ethernet LAN Switching (Part 1)

Ethernet involves L1 and L2 of OSI Model

LANs (Local Area Networks)

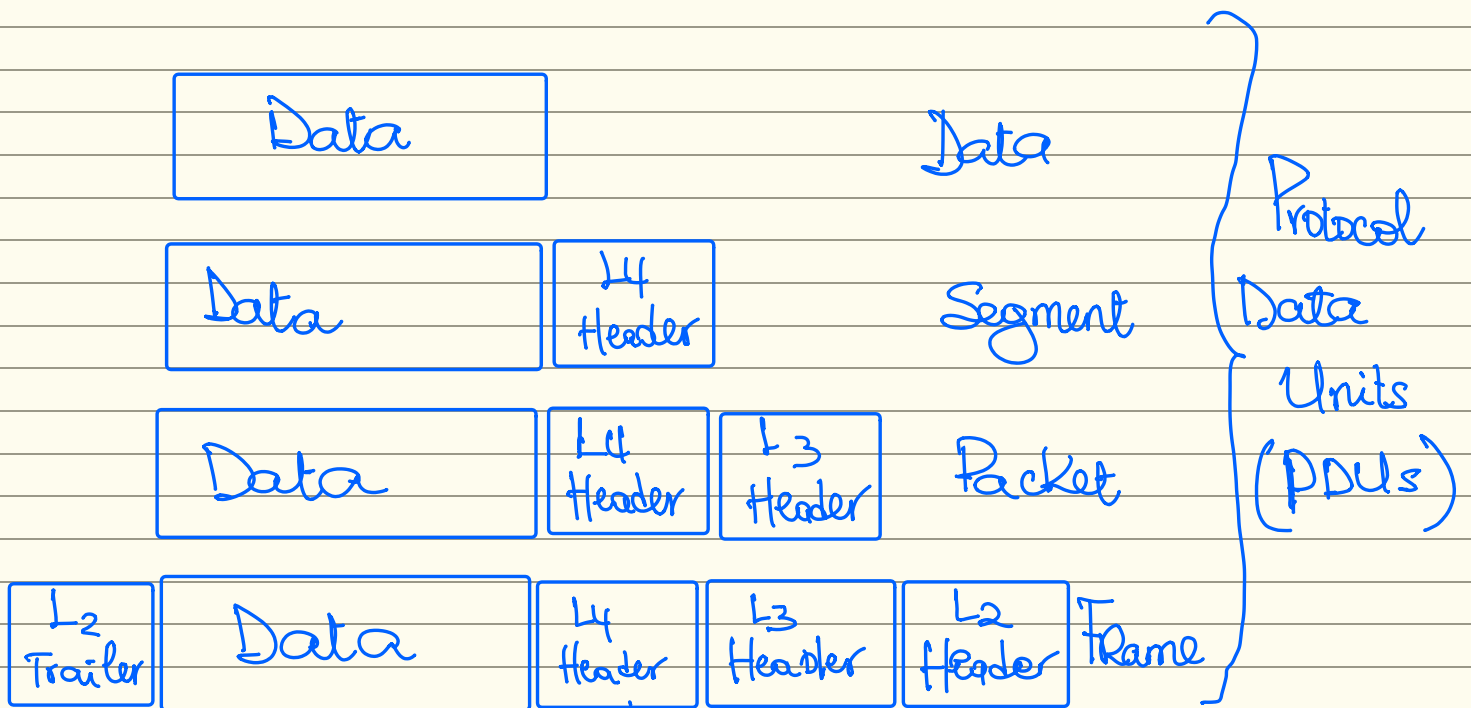
Network contained within a relatively small area, like an office floor or your home network.

* Switches do not separate LAN's, but adding more switches can be used to expand an existing LAN

L2 (Data Link)

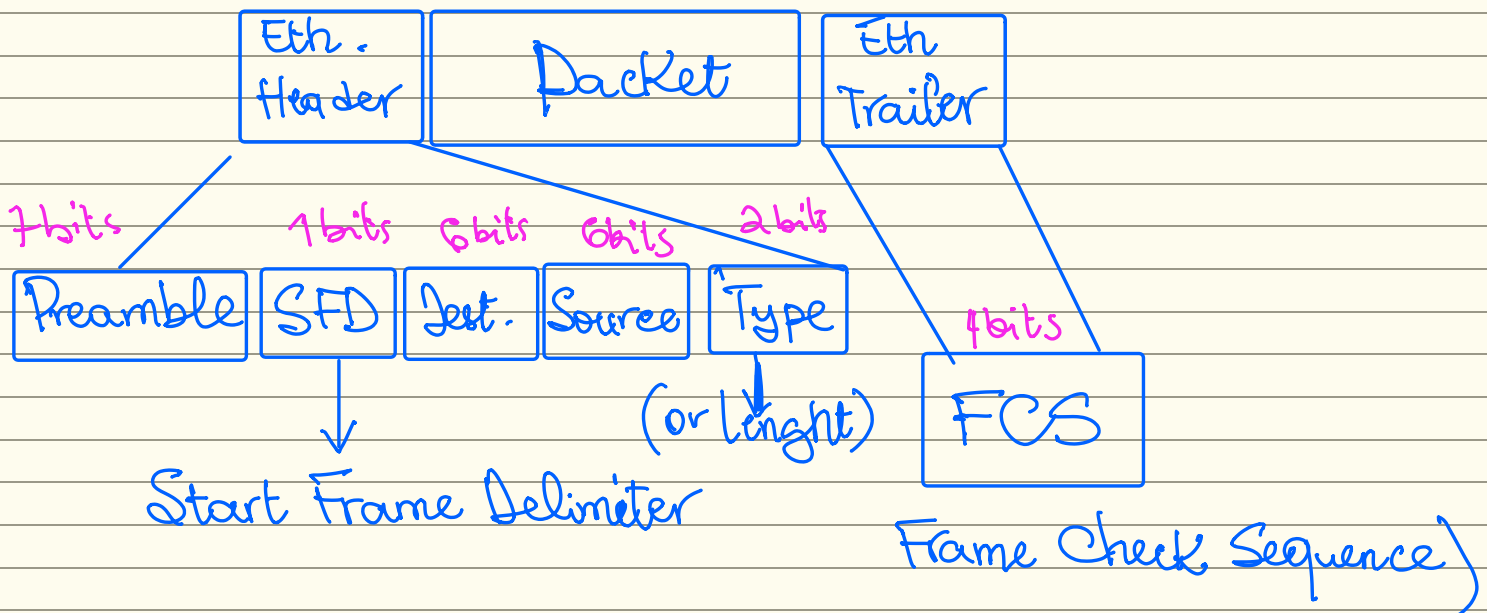
- Provides node-to-node connectivity and data transfer (for example, PC to switch, switch to router, router to router).
- Defines how data is formatted for transmission over a physical medium (for example, copper UTP cable).
- Detects and (possibly) corrects Physical Layer errors.
- Uses Layer 2 addressing, separate from Layer 3 addressing.

Switches operate at L2.



Eg. L2 PDU → Frame.

Ethernet Frame



Preamble & SFD

Preamble

- Length: 7 bytes (56 bits)
- Alternating 1's & 0's
- 10101010 * 7
- Allows devices to synchronize their receiver clocks

SFD

- Start Frame Delimiter
- Length: 1 byte (8 bits)
- 10101011
- Marks the end of the preamble, and the beginning of the rest of the frame.

Destination & Source

- Indicate the devices sending and receiving the frame
- Consist of the destination & source 'MAC address'
- MAC = Media Access Control
- 6 bytes (48 bits) address of the physical device

Type / Length

- 2 byte (16-bit) field
- A value of 1500 or less → length of the encapsulated packet (in bytes)
- A value of 1536 or greater → TYPE of the encapsulated packet (usually IPv4/IPv6), and the length is determined via other methods.

IPv4 = 0x0800 (hexo)
(2048 in decimal)

IPv6 = 0x086DD
(34525 in decimal)

Frame Check Sequence (FCS)

- 4 bytes (32 bits) in length
- Detects corrupted data by running a 'CRC' algorithm over the receiver data
- CRC = 'Cyclic Redundancy Check'

* Ethernet Frame's Frame Check Sequence is a Cyclic Redundancy Check.

MAC Address

- 6 bytes (48 bits) physical address assigned to the device when it is made
- A.K.A 'Burned-In Address' (BIA)
- Is globally unique
- The first 3 bytes are the OUI (Organizationally Unique Identifier), which is assigned to the company making the device.
- The last 3 bytes are unique to the device itself
- Written as 12 hexadecimal characters

Unicast Frame: a frame destined for a single target

Dynamic MAC Address: SW learned Source MAC Address → Table.

Unknown Unicast frame → FLOOD the frame

Known Unicast frame → destinatⁿ is already in SW MAC Address Table

Dynamic MAC Address are removed from the MAC address table after 5 minutes of inactivity.